

Algebra II with Trigonometry

Chapter 6.1

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4506904_alg-2trig-h-chp-6-1-note-docx.pdf](#)

- 1. Unit Circle Basics** What is the equation of the unit circle centered at the origin?
- 2. Starting Point** When measuring a real number t on the unit circle, from which point do you start?
- 3. Direction Check** If t is negative, do you move clockwise or counterclockwise around the unit circle?
- 4. Circle Distance** What is the total circumference of the unit circle?
- 5. Quarter Turn** What point on the unit circle is reached when $t = \frac{\pi}{2}$?
- 6. Half Turn** What point on the unit circle is reached when $t = \pi$?
- 7. Equivalent Angles** Do t and $t + 2\pi$ end at the same terminal point?
- 8. Reference Number** What is the reference number for $t = \frac{5\pi}{6}$?
- 9. Missing Coordinate** A point P is on the unit circle in Quadrant III and has $x = -\frac{3}{5}$. What is the y -coordinate?
- 10. Transforming a Terminal Point** If the terminal point for t is $(\frac{3}{5}, \frac{4}{5})$, what is the terminal point for $-t$?